Southern University Bangladesh
Department of Computer Science and Engineering

Silent Help+:University Anonymous Support & Resource Sharing Platform

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1. Introduction

1.1 Executive Summary

Silent Help+ is a comprehensive web-based platform designed to address critical challenges faced by university students. The system provides anonymous support channels, resource sharing, mentorship connections, and financial opportunities while maintaining strict privacy and department-wise organization. Built with modern web technologies (Node.js, MySQL, HTML/CSS), the platform serves as a safe space for students to seek help, share resources, and build supportive communities.

1.2 Problem Statement

University students face multiple interconnected challenges:

- Academic Struggles: Difficulty understanding concepts, fear of asking questions publicly
- Mental Health Issues: Bullying, harassment, lack of safe reporting mechanisms
- Financial Constraints: Limited access to scholarship and job opportunity information
- Resource Scarcity: Difficulty accessing study materials and expensive textbooks
- Limited Mentorship: Lack of structured peer guidance and career counseling

1.3 Proposed Solution

Silent Help+ addresses these challenges through a three-pillar approach:

Privacy Protection Through Controlled Anonymity: Students interact using system-generated anonymous IDs while admins maintain oversight capabilities for serious incidents.

Department-Wise Content Organization: Academic content (Q&A, resources) filtered by department ensuring relevance while preventing information overload.

University-Wide Common Resources: Scholarships, jobs, and general opportunities accessible to all students regardless of department.

1.4 Core Features Overview

1)Anonymous Help Request System
2)Resource Library (Study Materials)
3)Anonymous Reporting System
4)Peer Mentorship Matching
5)Book Exchange Platform
6)Scholarship Hub (University-Wide)
7)Job Board (University-Wide)

2. Core Features

2.1 Anonymous Help Request System

Department-wise Q&A platform where students post questions anonymously. System generates unique anonymous IDs for each user. Other students from the same department can view and answer questions. Voting system ranks answers by quality.

Privacy Mechanism: Full anonymity with department isolation

2.2 Resource Library

Repository for study materials (lecture notes, past papers, assignments). Students upload files with metadata (subject, semester, description). Admin approval system ensures quality. Search and filter functionality by subject and semester.

Privacy Mechanism: Approval workflow, department filtering

2.3 Anonymous Reporting System

Confidential reporting channel for harassment, bullying, discrimination, and safety concerns. Reports submitted with anonymous reporter IDs. Admin can review, investigate, and trace identity only for serious cases. All tracing actions are logged for accountability.

Privacy Mechanism: Complete anonymity with admin oversight

2.4 Peer Mentorship Matching

Connects junior students with senior mentors based on expertise and interests. Mentors register with their skills, availability, and areas of guidance. Mentees search and send connection requests. Optional transition to direct contact after mutual agreement.

Privacy Mechanism: Profile-based matching, optional anonymity

2.5 Book Exchange Platform

Marketplace for textbooks - students can sell, rent, or exchange books. Listings include book details, condition, price, and exchange type. Direct communication between students for transaction arrangement. Owner manages listing status (available, sold, rented).

Privacy Mechanism: Direct peer-to-peer exchange

2.6 Scholarship Hub

Centralized database of scholarship opportunities accessible to ALL students regardless of department. Listings include eligibility criteria, amount, application deadline. Auto-expiration system for past deadlines. Filter by amount, deadline, and eligibility.

Privacy Mechanism: University-wide access, no department restriction

2.7 Job Board

Part-time jobs, internships, and full-time positions accessible to ALL students. Companies or admin post listings with requirements and salary. Filter by job type, location, and salary range. Application links and contact information provided. Auto-expiration after 60 days.

Privacy Mechanism: University-wide access, no department restriction

3. System Architecture

3.1 Technology Stack

Frontend	**HTML5, CSS3, Bootstrap 5, JavaScript (ES6)**
Backend	Node.js, Express.js
Database	MySQL
Security	JWT (Authentication), bcrypt (Password Hashing)
Additional	Nodemailer (Email), Multer (File Upload)

3.2 Three-Tier Architecture

Presentation Layer (Frontend): User interface components built with HTML/CSS/JavaScript. Bootstrap ensures responsive design across devices. Client-side validation and dynamic content loading.

Application Layer (Backend): Node.js server with Express.js routing. Handles authentication, business logic, API endpoints. Processes requests and enforces security rules. Manages file uploads and email notifications.

Data Layer (Database): MySQL database stores all application data. Nine core tables with foreign key relationships. Ensures data integrity and supports complex queries for filtering and searching.

4. Database Design

4.1 Database Tables

Table Name	Key Fields	Purpose
users	id, name, email, password, department, student_id, is_verified, created_at	Core user authentication and profile
help_requests	id, user_id, anonymous_id, department, category, question_text, created_at	Anonymous Q&A with department filtering
answers	id, request_id, answerer_id, anonymous_id, answer_text, votes, created_at	Answers with voting system
resources	id, uploader_id, file_path, subject, semester, department, status, created_at	Study materials with approval
reports	id, reporter_id, anonymous_id, report_type, description, status, created_at	Anonymous incident reports
mentors	id, user_id, expertise, availability, bio, is_active, created_at	Mentor profiles for guidance
scholarships	id, title, amount, eligibility, deadline, description, created_at	University-wide scholarships
job_posts	id, title, company, job_type, salary, location, deadline, created_at	University-wide job listings
book_exchange	id, owner_id, book_title, author, condition, price, status, created_at	Textbook marketplace

4.2 Key Relationships

- users (1) → (N) help_requests: One user posts many questions
- users (1) → (N) answers: One user provides many answers
- users (1) → (N) resources: One user uploads many resources
- users (1) → (1) mentors: One user registers as one mentor
- help_requests (1) → (N) answers: One question receives many answers

5. Security & Privacy Measures

5.1 Authentication System

- JWT (JSON Web Token) for stateless session management
- bcrypt password hashing with salt rounds for secure storage
- Email verification system using unique verification tokens
- Session timeout and automatic logout after inactivity
- Password strength requirements enforced during registration

5.2 Controlled Anonymity System

The platform implements a dual-layer identity system:

- Real Identity Layer: System knows user_id from authenticated session
- Anonymous Identity Layer: System generates unique anonymous_id for each interaction
- Public Display: Only anonymous_id shown to other users
- Admin Access: Can trace anonymous_id to user_id for serious violations
- Audit Trail: All identity tracing actions logged with timestamp and reason

5.3 Department Isolation

Academic features implement strict department filtering:

- Help requests visible only to same department students
- Resource library filtered by department automatically
- SQL queries include WHERE department = user.department clause
- Cross-department access prevented at database level
- Scholarships and jobs remain university-wide (no restriction)

5.4 Input Validation & Security

- Server-side validation for all user inputs
- SQL injection prevention using parameterized queries
- XSS (Cross-Site Scripting) protection through input sanitization
- File upload validation (type, size, extension whitelist)
- Rate limiting to prevent spam and abuse

6. Implementation Plan

6.1 Timeline (4 Months)

Month 1: Foundation & Core Setup

- Install development tools: Node.js, MySQL, VS Code, Git
- Design complete database schema with all 9 tables
- Create user authentication system (register, login, email verification)
- Build basic dashboard interface

Month 2: Core Features Development

- Implement anonymous help request system with voting
- Build resource library with upload/download functionality
- Add department-based filtering to all academic features
- Create admin approval workflow for resources

Month 3: Additional Features & Admin Panel

- Develop scholarship hub and job board (university-wide)
- Build book exchange platform
- Implement peer mentorship matching system
- Create anonymous reporting system
- Build comprehensive admin dashboard

Month 4: Testing & Documentation

- Comprehensive testing (unit, integration, security)
- Bug fixing and performance optimization
- Write user manual and technical documentation
- Prepare presentation and demo scenarios

6.2 Team Responsibilities

Member 1: Frontend Developer: Responsible for UI/UX design, HTML/CSS/Bootstrap implementation, JavaScript interactivity, responsive design, and user interface testing.

Member 2: Backend Developer: Handles Node.js/Express.js server setup, API endpoints, authentication system, business logic, email verification, and file upload processing.

Member 3: Database & Integration Specialist: Manages MySQL database design, SQL query optimization, frontend-backend integration, end-to-end testing, and data integrity.

6.3 Admin Control Features

- User Management: View all users, verify student IDs, ban accounts, reset passwords
- Content Moderation: Review and delete inappropriate posts, approve study materials
- Report Management: Investigate anonymous reports, trace identities when needed
- System Analytics: User statistics, activity trends, popular topics, resolution metrics

7. Risk Analysis & Expected Outcomes

7.1 Risk Analysis

Risk	Mitigation Strategy
Privacy Breach	Strong encryption, secure JWT tokens, audit logging for all admin actions
Spam/Abuse	Rate limiting, reporting system, admin moderation, account bans
Data Loss	Regular database backups, transaction logging, data redundancy
Performance Issues	Query optimization, database indexing, caching strategies
Unauthorized Access	Multi-layer authentication, session management, password policies

7.2 Expected Outcomes

Quantifiable Metrics

- Support 500+ concurrent users without performance degradation
- Handle 1000+ questions posted per month across departments
- Store 500+ study resources in organized library
- Facilitate 50+ active mentor-mentee connections
- List 100+ scholarship and job opportunities
- Page load time under 2 seconds
- System uptime of 99%+

Benefits for Stakeholders

For Students: Anonymous help without judgment, access to curated study materials, financial opportunities, mentorship connections, safe reporting channel, cost savings through book exchange.

For University: Improved student satisfaction, better resource sharing, early detection of campus issues, enhanced safety culture, data-driven policy decisions, positive institutional reputation.

For Society: Replicable model for other universities, promotes mental health awareness, demonstrates technology for social good, builds mutual support culture.

8. Conclusion

8.1 Summary

Silent Help+ represents a comprehensive solution to the multifaceted challenges faced by university students. By combining anonymous support mechanisms, resource sharing platforms, and financial opportunity hubs, the system creates a holistic ecosystem that addresses academic, social, and economic needs.

The platform's unique three-pillar approach—controlled anonymity, department-wise organization, and university-wide resources—ensures both privacy and relevance. The technical architecture, built on proven technologies like Node.js and MySQL, provides a robust, scalable foundation capable of supporting hundreds of concurrent users while maintaining high performance and security standards.

8.2 Key Achievements

- Seven integrated features addressing different aspects of student life
- Privacy-preserving anonymous interaction system
- Department-based content filtering for relevance
- Comprehensive admin control with accountability
- Scalable three-tier architecture
- Robust security implementation

8.3 Future Scope

- Mobile Application: Native Android/iOS apps for better accessibility
- AI-Powered Features: Intelligent answer suggestions, automatic content categorization
- Multi-University Expansion: Extend platform to multiple institutions with inter-university resource sharing
- LMS Integration: Connect with existing Learning Management Systems
- Analytics Dashboard: Advanced reporting and visualization for administrators
- Notification System: Real-time alerts for mentions, answers, and opportunities
- Gamification: Reputation points and badges for helpful contributors